

Structure of Marking Scheme

Scale label	A	B	C	D
No of categories	2	3	4	5
5- markscale	0, 5	0, 3, 5	0, 2, 3, 5	0, 2, 3, 4, 5
10- markscale			0, 4, 7, 10	0, 3, 5, 8, 10
15- markscale				0, 4, 8, 12, 15

A general descriptor of each point on each scale is given below. More specific directions in relation to interpreting the scales in the context of each question are given in the body of the scheme, where necessary.

Markingscales–level descriptors

A-scales (two categories)

- incorrect response (no credit)
- correct response (full credit)

B-scales (three categories)

- response of no substantial merit (no credit)
- partially correct response (partial credit)
- correct response (full credit)

C-scales (four categories)

- response of no substantial merit (no credit)
- response with some merit (low partial credit)
- almost correct response (high partial credit)
- correct response (full credit)

D-scales (five categories)

- response of no substantial merit (no credit)
- response with some merit (low partial credit)
- response about half-right (middle partial credit)
- almost correct response (high partial credit)
- correct response (full credit)

In certain cases, typically involving incorrect rounding, omission of units, a misreading that does not oversimplify the work, or an arithmetical error that does not oversimplify the work, a mark that is one mark below the full-credit mark may also be awarded. Such cases are denoted with a * and this level of credit is referred to as full credit -1. Thus, for example, in Scale 10C, full credit -1 of 9 marks may be awarded.

The only marks that may be awarded for a question are those on the scale above, or *Full Credit -1*. A rounding penalty is applied only once in each section (a), (b), (c) etc. A penalty for an omitted unit is applied only once in each section (a), (b), (c) etc. There is no penalty for omitted units if the question specifies the unit to be used in the answer, and there is generally no penalty for an omitted euro symbol in questions involving money.

In general, accept a candidate's work in one part of a question for use in subsequent parts of the question, unless this oversimplifies the work involved.

Detailed marking notes

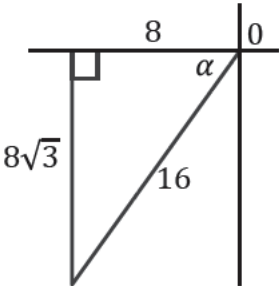
Model Solutions & Marking Notes

Note: The model solutions for each question are not intended to be exhaustive – there may be other correct solutions.

Q1	Model Solution – 30 Marks	Marking Notes
(a)	$a^2 + 2ab + b^2 - 49 = (a+b)^2 - 7^2$ $= (a+b+7)(a+b-7)$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> - Work of merit, for example, a common factor identified <i>High Partial Credit:</i> $(a+b)^2 - 7^2$ $a(a+2b) + (b-7)(b+7)$
(b)	$3m - 8n - 2 + 12mn = 12mn + 3m - 8n - 2$ $= 3m(4n+1) - 2(4n+1)$ $= (3m-2)(4n+1)$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> - Work of merit, for example, a common factor identified <i>High Partial Credit:</i> $3m(4n+1) - 2(4n+1)$
(c)	$\frac{2}{x} - \frac{2y}{x} = 1 \Rightarrow \frac{2-2y}{x} = 1$ $2-2y = x \quad x^2 + 2xy - 8 = 0$ $(2-2y)^2 + 2(2-2y)y - 8 = 0$ $4-8y+4y^2+4y-4y^2-8=0$ $-4y-4=0 \Rightarrow y=-1$ $x=2-2(-1) \Rightarrow x=4$ or $\frac{2}{x} - \frac{2y}{x} = 1 \Rightarrow \frac{2-2y}{x} = 1$ $2-2y = x \Rightarrow 2y = 2-x$ $x^2 + 2xy - 8 = 0$ $x^2 + (2-x)x - 8 = 0$ $x^2 + 2x - x^2 - 8 = 0$ $2x - 8 = 0 \Rightarrow x = 4$ $y = \frac{2-x}{2} \Rightarrow y = \frac{2-4}{2} \Rightarrow y = -1$ (4, -1)	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, effort to isolate x or y <i>Mid Partial Credit:</i> - Finds $(2-2y)^2 + 2y(2-2y) - 8 = 0$ - Finds $x^2 + (2-x)x - 8 = 0$ <i>High Partial Credit:</i> - Finds correct value for x or y

Q2	Model Solution – 30 Marks	Marking Notes
(a) & (b)	(a) $f(5) = 5 - \frac{1}{e^5}$ $= 4.993262053$ $= 4.9933$ (b) $\lim_{x \rightarrow \infty} \left(5 - \frac{1}{e^x} \right) = 5 - 0 = 5$	Scale 10C (0, 4, 7, 10) <i>Note:</i> Accept correct answers with no work. <i>Low Partial Credit:</i> - Work of merit, for example, any correct substitution <i>High Partial Credit:</i> (a) correct and work of merit in (b)
(c)	$5 - \frac{1}{e^x} = \frac{1}{5}$ $\frac{1}{e^x} = 5 - \frac{1}{5}$ $\frac{1}{e^x} = \frac{25-1}{5} = \frac{24}{5}$ $e^x = \frac{5}{24}$ $x = \ln \frac{5}{24} = \ln 5 - \ln 24$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, sets up equation <i>Mid Partial Credit:</i> - Isolates e^x <i>High Partial Credit:</i> - Finds $x = \ln \frac{5}{24}$
(d)	$\log_5(x+2) + \log_5(x-2) = 5$ $\log_2(x+2)(x-2) = 5 \log_2 2$ $\log_2(x^2 - 4) = \log_2 2^5$ $x^2 - 4 = 32$ $x^2 = 32 + 4$ $x^2 = 36$ $x = \pm 6$ $\therefore x = 6 \text{ as } x = -6 \text{ not valid}$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> - Work of merit, for example, any correct use of logs <i>High Partial Credit:</i> - Correct quadratic equation <i>Full credit – 1</i> - Fails to reject $x = -6$

Q3	Model Solution – 30 Marks	Marking Notes
(a)	$\frac{5+2i}{2-5i} \times \frac{2+5i}{2+5i} = \frac{10+25i+4i+10i^2}{4+10i-10i-25i^2}$ $= \frac{10+29i-10}{4+25}$ $= \frac{29i}{29} = i$ Thus $\left(\frac{5+2i}{2-5i}\right)^5 = i^5 = i^2 \times i^2 \times i = i$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> • Work of merit, for example, states the conjugate of either complex number <i>Mid Partial Credit:</i> • Multiplies top and bottom by the conjugate of the denominator <i>High Partial Credit:</i> • Simplifies to i, but fails to correctly find i^5
(b)	$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$ [de Moivre's Thm] $(\cos \theta + i \sin \theta)^3$ using a Binomial expansion: $= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$ $= \cos^3 \theta + 3i \cos^2 \theta \sin \theta + 3i^2 \cos \theta \sin^2 \theta + i^3 \sin^3 \theta$ $= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$ $= \cos^3 \theta - 3 \cos \theta \sin^2 \theta + 3i \cos^2 \theta \sin \theta - i \sin^3 \theta$ Equating imaginary parts: $\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$ $= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta$ $= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta$ $= 3 \sin \theta - 4 \sin^3 \theta$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit</i> • Work of merit, for example, use of de Moivre's theorem or Binomial expansion <i>Mid Partial Credit</i> • Fully correct application of de Moivre's theorem or Binomial expansion <i>High Partial Credit</i> • $\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$

Q3	Model Solution – 30 Marks	Marking Notes
(c)	$r = -8 - 8\sqrt{3}i $ $r = \sqrt{(-8)^2 + (-8\sqrt{3})^2}$ $= \sqrt{64 + 192}$ $= \sqrt{256}$ $= 16$ $\tan \alpha = \frac{8\sqrt{3}}{8} = \sqrt{3}$ $\alpha = 60^\circ$ $\theta = 180^\circ + 60^\circ = 240^\circ$ $z^4 = -8 - 8\sqrt{3}i$ $z^4 = 16(\cos 240^\circ + i \sin 240^\circ)$ $z = [16(\cos 240^\circ + i \sin 240^\circ)]^{\frac{1}{4}}$ $z = [16(\cos(240^\circ + 360^\circ n) + i \sin(240^\circ + 360^\circ n))]^{\frac{1}{4}}$ $= 16^{\frac{1}{4}} \left[\cos \frac{1}{4}(240^\circ + 360^\circ n) + i \sin \frac{1}{4}(240^\circ + 360^\circ n) \right]$ $= 2[\cos(60^\circ + 90^\circ n) + i \sin(60^\circ + 90^\circ n)]$ When $n = 0$ $z = 2[\cos 60^\circ + i \sin 60^\circ] = 2\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = 1 + \sqrt{3}i$ When $n = 1$ $z = 2[\cos 150^\circ + i \sin 150^\circ] = 2\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = -\sqrt{3} + i$ When $n = 2$ $z = 2[\cos 240^\circ + i \sin 240^\circ] = 2\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = -1 - \sqrt{3}i$ When $n = 3$ $z = 2[\cos 330^\circ + i \sin 330^\circ] = 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = \sqrt{3} - i$ 	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, finds modulus or argument or shows some knowledge of De Moivre's theorem <i>Mid Partial Credit:</i> - Finds $16[\cos(240^\circ) + i \sin(240^\circ)]$ <i>High Partial Credit:</i> - One root found in the form $a + bi$ from De Moivre's expression

Q4	Model Solution – 30 Marks	Marking Notes
(a)	$u = x^2 - 1 \quad v = x^2 + 1$ $\frac{du}{dx} = 2x \quad \frac{dv}{dx} = 2x$ $\frac{dy}{dx} = \frac{2x(x^2 + 1) - 2x(x^2 - 1)}{(x^2 + 1)^2}$ $= \frac{2x^3 + 2x - 2x^3 + 2x}{(x^2 + 1)^2}$ $= \frac{4x}{(x^2 + 1)^2}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, any correct differentiation <i>Mid Partial Credit:</i> - Any correct use of Quotient rule <i>High Partial Credit:</i> - Fully correct substitution into Quotient rule
(b)	$g'(x) = 3e^x - \cos x$ (i) $g'(0) = 3e^0 - \cos 0$ & $= 3 - 1 = 2$ (ii) $g(0) = 3e^0 - \sin 0 + 5$ $= 3 - 0 + 5 = 8$ $m = 2$ Point $= (0, 8)$ $y - y_1 = m(x - x_1)$ $y - 8 = 2(x - 0)$ $y - 8 = 2x$ $2x - y + 8 = 0$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, any correct differentiation <i>Mid Partial Credit:</i> - Part (i) correct <i>High Partial Credit:</i> - Finds correct slope and point
(c)	$y = x^2 \tan x$ (i) $u = x^2 \quad v = \tan x$ $\frac{du}{dx} = 2x \quad \frac{dv}{dx} = \sec^2 x$ $\frac{dy}{dx} = x^2 \sec^2 x + 2x \tan x$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit</i> - Work of merit, for example, any correct differentiation <i>High Partial Credit:</i> - Finds both $\frac{du}{dx}$ and $\frac{dv}{dx}$
(c)	(ii) $\int (x \tan x + 1)^2 dx = \int (x^2 \tan^2 x + 2x \tan x + 1) dx$ $= \int (x^2 (\sec^2 x - 1) + 2x \tan x + 1) dx$ $= \int (x^2 \sec^2 x - x^2 + 2x \tan x + 1) dx$ $= \int (x^2 \sec^2 x + 2x \tan x - x^2 + 1) dx$ $= \int (x^2 \sec^2 x + 2x \tan x) dx + \int (-x^2 + 1) dx$ $= x^2 \tan x - \frac{x^3}{3} + x + c$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit</i> - Work of merit, for example, any correct term in expansion of brackets <i>High Partial Credit:</i> - Any correct integration

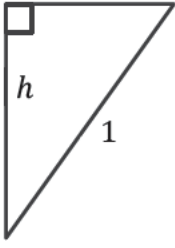
Q5	Model Solution – 30 Marks	Marking Notes
(a)	$x^2 - x(\text{sum of roots}) + \text{product of roots} = 0$ $x^2 - x\left(\frac{5}{2} + \frac{3}{5}\right) + \left(\frac{5}{2} \times \frac{3}{5}\right) = 0$ $x^2 - x\left(\frac{31}{10}\right) + \left(\frac{3}{2}\right) = 0$ $10x^2 - 31x + 15 = 0$ or Let $x = \frac{5}{2}$ and let $x = \frac{3}{5}$ $2x = 5 \quad 5x = 3$ $2x - 5 = 0 \quad 5x - 3 = 0$ $(2x - 5)(5x - 3) = 0$ $10x^2 - 6x - 25x + 15 = 0$ $10x^2 - 31x + 15 = 0$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, attempts to find the sum or product of the roots <i>Mid Partial Credit:</i> - One correct factor found - Sum or product of roots found <i>High Partial Credit:</i> - One of a, b or c found <i>Full Credit –1:</i> $10x^2 - 31x + 15$
(b) (i)	$2x^2 + 4x - 5 = 2(x^2 + 2x) - 5$ $= 2(x^2 + 2x + 1 - 1) - 5$ $= 2(x + 1)^2 - 2 - 5$ $= 2(x + 1)^2 - 7$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, $a = 2$ identified explicitly or as a factor <i>Mid Partial Credit:</i> - Completed square found <i>High Partial Credit:</i> - One of h or k found
(b) (ii)	The minimum value occurs when $(x + 1) = 0$ $\therefore$ minimum point is $(-1, -7)$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit</i> - Work of merit, for example, states the minimum occurs when $(x + 1) = 0$ <i>High Partial Credit:</i> - Finds correct point but no justification - Uses calculus to find correct point
(c)	The point is a minimum as $a > 0$ and as the y -coordinate is < 0 , the graph of $f(x)$ must cut the x -axis twice, $\therefore$ there are 2 real roots or $b^2 - 4ac = 4^2 - 4(2)(-5)$ $= 16 + 40 = 56 > 0$ $\therefore$ there are 2 real roots	Scale 5B (0, 3, 5) <i>Partial Credit</i> Work of merit, for example, states a is positive or $b^2 - 4ac > 0$

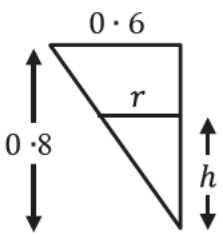
Q6	Model Solution – 30 Marks	Marking Notes
(a)	$g'(x) = 3x^2 - 12x + 9$	Scale 15D (0, 4, 8, 12, 15)
(i)	$g''(x) = 6x - 12$	4 steps
(ii)	$g'(x) = 0$ at local maximum/minimum	1. Finds $g'(x)$
	$3x^2 - 12x + 9 = 0$	2. Solves $g'(x) = 0$
	$x^2 - 4x + 3 = 0$	3. Solves $g''(x) = 0$
	$(x - 1)(x - 3) = 0$	4. Finds the coordinates of the 3 points
	$x = 1$ and $x = 3$	
	When $x = 1$	<i>Low Partial Credit:</i>
	$g''(1) = 6(1) - 12$	• Work of merit, for example, any correct differentiation
	$= -6 < 0 \Rightarrow$ maximum	
	$g(1) = 1^3 - 6(1)^2 + 9(1) - 5$	<i>Mid Partial Credit:</i>
	$= 1 - 6 + 9 - 5 = -1$	• 2 steps correct
	Maximum at $(1, -1)$	• Finds the coordinates of any one of the local maximum, local minimum or point of inflection
	When $x = 3$	
	$g''(3) = 6(3) - 12$	<i>High Partial Credit:</i>
	$= 6 > 0 \Rightarrow$ minimum	• 3 steps correct
	$g(3) = 3^3 - 6(3)^2 + 9(3) - 5$	• Finds the coordinates of any two of the local maximum, local minimum or point of inflection
	$= 27 - 54 + 27 - 5 = -5$	
	Minimum at $(3, -5)$	
	$g''(x) = 0$ at point of inflection	
	$6x - 12 = 0 \Rightarrow x = 2$	
	$g(2) = 2^3 - 6(2)^2 + 9(2) - 5$	
	$= 8 - 24 + 18 - 5 = -3$	
	Point of inflection at $(2, -3)$	
	Local minimum : $(3, -5)$	
	Local maximum : $(1, -1)$	
	Point of inflection: $(2, -3)$	

Q6	Model Solution – 30 Marks	Marking Notes
(b)	$ \begin{aligned} f(x+h) - f(x) &= 5(x+h)^2 - 7(x+h) - (5x^2 - 7x) \\ &= 5(x^2 + 2xh + h^2) - 7x - 7h - 5x^2 + 7x \\ &= 5x^2 + 10xh + 5h^2 - 7x - 7h - 5x^2 + 7x \\ \frac{f(x+h) - f(x)}{h} &= \frac{10xh + 5h^2 - 7h}{h} \\ \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} &= 10x + 5h - 7 \\ &= 10x - 7 \\ f'(x) &= 10x - 7 \end{aligned} $	Scale 15D (0, 4, 8, 12, 15) <i>Note:</i> no credit unless differentiation from first principles is being used. 4 steps: 1. Finds $f(x+h)$ 2. Finds $f(x+h) - f(x)$ 3. Finds $\frac{f(x+h) - f(x)}{h}$ 4. Finds $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ <i>Low Partial Credit:</i> • Work of merit, for example, some substitution into $f(x+h)$ <i>Mid Partial Credit:</i> • 2 steps correct <i>High Partial Credit:</i> • 3 steps correct • Left-hand side missing, otherwise correct

Q7	Model Solution – 50 Marks	Marking Notes
(a)	Initial population when $t = 0$ $B(0) = 4 \cdot 5e^{1.4(0)}$ $= 4 \cdot 5$ Bees = $4 \cdot 5 \times 1000$ $= 4500$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - States or implies $t = 0$ <i>High Partial Credit:</i> - Finds $B(0) = 4 \cdot 5e^{1.4(0)}$
(b)	$B(0.5) = 4 \cdot 5e^{1.4(0.5)}$ $= 4 \cdot 5e^{0.7}$ $= 9 \cdot 061887184$ Bees = $9 \cdot 061887 \times 1000$ Increase = $9061 \cdot 887 - 4500$ $= 4561 \text{ (also accept 4562)}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, any correct substitution into formula <i>High Partial Credit:</i> - Finds $9 \cdot 061$ or equivalent
(c)	$B'(t)$ is the rate of growth $B'(t) = 4 \cdot 5e^{1.4t} \times 1.4$ $= 6 \cdot 3e^{1.4t}$ $B'(2) = 6 \cdot 3e^{1.4(2)}$ $= 6 \cdot 3e^{2.8}$ $= 103 \cdot 6012 \dots$ Rate of growth is 103,601 bees per year	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> - Work of merit, for example, any correct differentiation <i>High Partial Credit:</i> - Finds $B'(t) = 6 \cdot 3e^{1.4t}$
(d)	$B'(t) = 6 \cdot 3e^{1.4t}$ $6 \cdot 3e^{1.4t} = 60$ $e^{1.4t} = \frac{60}{6 \cdot 3}$ $1.4t = \ln\left(\frac{60}{6 \cdot 3}\right)$ $t = \frac{\ln\left(\frac{60}{6 \cdot 3}\right)}{1.4}$ $= 1 \cdot 6098 \dots$ $= 1 \cdot 61 \text{ years}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, any correct substitution into equation <i>Mid Partial Credit:</i> - Fully substituted equation <i>High Partial Credit:</i> - Equation in t with no indices (logs handled correctly)
(e) (i)	$B(3) = 4 \cdot 5e^{1.4(3)}$ $= 4 \cdot 5e^{4.2} \times 1000$ $= 300,088 \cdot 4897$ Population of bees = 300,088 $Ae^{k(0)} = 300$ $A = 300$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, any correct substitution into formula <i>High Partial Credit:</i> - Finds population of bees is 300,088

Q7	Model Solution – 50 Marks	Marking Notes																
(e)	$300e^{k(1)} = 180$	Scale 5C (0, 2, 3, 5)																
(ii)	$e^k = \frac{180}{300} = \frac{3}{5} = 0.6$ $k = \ln(0.6)$ $= -0.5108256238$ $= -0.5108$	<i>Low Partial Credit:</i> - Work of merit, for example, any correct substitution into formula <i>High Partial Credit:</i> - Finds $e^k = \frac{180}{300}$																
(f)	<table><tr><th>t (year)</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr><tr><td>$b(t) = 300e^{-0.5108t}$</td><td>300</td><td>180</td><td>108</td><td>64.8</td><td>38.9</td><td>23.3</td><td>14.0</td></tr></table> Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Two points found correctly - One point plotted correctly <i>Mid Partial Credit:</i> - Four points plotted correctly <i>High Partial Credit:</i> - Five points plotted correctly <i>Full Credit – 1:</i> - All points plotted correctly but not joined or joined inappropriately - Consistent incorrect rounding		t (year)	0	1	2	3	4	5	6	$b(t) = 300e^{-0.5108t}$	300	180	108	64.8	38.9	23.3	14.0
t (year)	0	1	2	3	4	5	6											
$b(t) = 300e^{-0.5108t}$	300	180	108	64.8	38.9	23.3	14.0											

Q8	Model Solution – 50 Marks	Marking Notes
(a) Diameter = $1.2 \Rightarrow r = 0.6$ (i) Pythagoras' theorem: $h^2 + 0.6^2 = 1^2$ $h^2 = 1 - 0.36$ $= 0.64$ $h = \sqrt{0.64}$ $= 0.8 \text{ metres}$		Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, finds the radius or draws relevant diagram <i>High Partial Credit:</i> - Any correct substitution into Pythagoras' theorem
(a) Area of cylinder = $2\pi rh + \pi r^2$ (ii) $= 2\pi(0.6)(2) + \pi(0.6)^2$ $= 2.76\pi = 8.6707\dots$ Area of cone = πrl $= \pi(0.6)(1)$ $= 0.6\pi = 1.8849\dots$ Total area = $0.6\pi + 2.76\pi$ $= 3.36\pi$ $= 10.55575132$ Number of cans = $\frac{10.56}{3}$ $= 3.518\dots$ $= 4 \text{ cans}$		Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, any correct substitution into area of cylinder or cone or circle <i>Mid Partial Credit</i> - Finds correct area of cone or cylinder <i>High Partial Credit</i> - Finds correct total area
(a) Volume of cylinder = $\pi r^2 h$ (iii) $= \pi(0.6)^2(2)$ $= 2.261946711$ Volume of cone = $\frac{1}{3}\pi r^2 h$ $= \frac{1}{3}\pi(0.6)^2(0.8)$ $= 0.3015928947$ Total = $2.26 + 0.30$ $= 2.56 \text{ [m}^3\text{]}$		Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, any correct substitution into volume of cylinder or cone <i>Mid Partial Credit:</i> - Any correct substitution into both volume formulae <i>High Partial Credit:</i> - Fully correct substitution into both volume formulae

Q8	Model Solution – 50 Marks	Marking Notes
(b) (i)	Using similar triangles  $\frac{r}{h} = \frac{0.6}{0.8}$ $= 0.75$ $r = 0.75h$	Scale 5B (0, 3, 5) <i>Partial Credit:</i> Work of merit, for example, any relevant diagram or use of similar triangles
(b) (ii)	$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi (0.75h)^2 h = \frac{3}{16}\pi h^3$ $\frac{dV}{dh} = \frac{3}{16}3\pi h^2 = 0.5625\pi h^2$ $\frac{dh}{dV} = \frac{1}{0.5625\pi h^2} \quad \frac{dV}{dt} = 1.2$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $= \frac{1}{0.5625\pi h^2} \times 1.2$ $= \frac{1.2}{0.5625\pi h^2}$ When $h = 0.5$ $\frac{dh}{dt} = \frac{1.2}{0.5625\pi (0.5)^2}$ $= 2.716244362$ $= 2.72 \text{ [m/min]}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, states a relevant derivative or any correct differentiation <i>Mid Partial Credit:</i> - Any correct substitution into $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ or equivalent <i>High Partial Credit:</i> - Finds $\frac{dh}{dt} = \frac{1}{0.5625\pi h^2} \times 1.2$
(c)	$r = 0.75h \Rightarrow h = \frac{r}{0.75}$ $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi r^2 \left(\frac{r}{0.75}\right) = \frac{4\pi r^3}{9}$ $\frac{dV}{dr} = \frac{4\pi r^2}{3} \quad \frac{dr}{dt} = 0.04$ $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ $= \frac{4\pi r^2}{3} \times 0.04$ $= \frac{4\pi r^2}{75}$ When $r = 0.5$ $\frac{dV}{dt} = \frac{4\pi (0.5)^2}{75}$ $= 0.0418879...$ $= 0.042 \text{ [m}^3\text{/min]}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, finds $h = \frac{r}{0.75}$ or any correct differentiation <i>Mid Partial Credit:</i> - Any correct substitution into $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ or equivalent <i>High Partial Credit:</i> - Finds $\frac{dV}{dt} = \frac{2\pi (0.5)^2}{2.25}$

Q9	Model Solution – 50 Marks	Marking Notes
(a) (i)	$20 \times 0.7^5 = 3.3614$ $= 3.36 \text{ [mg]}$	Scale 5B (0, 2, 5) <i>Note:</i> Accept correct answer without work. <i>Low Partial Credit</i> Work of merit, for example, correct substitution into given expression
(a) (ii)	$20(0.7)^n < 1$ $0.7^n < \frac{1}{20} = 0.05$ $n \log 0.7 < \log 0.05$ $n > \frac{\log 0.05}{\log 0.7}$ $n > 8.39905\dots$ The amount of drug will be less than 1 mg after 9 days.	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, $20(0.7)^n < 1$ or finds correct answer by trial and improvement <i>Mid Partial Credit:</i> - Finds $0.7^n < 0.05$ or equivalent <i>High Partial Credit:</i> - Equation in n with no indices (logs handled correctly)
(a) (iii) & (iv)	There is 20 mg from the tablet just taken, and the amount from each previous tablet has reduced by 30% each day. or 20 mg is from the tablet just taken $20(0.7)$ is from the tablet taken one day ago $20(0.7)^2$ is from the tablet taken two days ago $20(0.7)^3$ is from the tablet taken three days ago $20(0.7)^4$ is from the tablet taken four days ago $20 + 20(0.7) + 20(0.7)^2 + 20(0.7)^3 + \dots + 20(0.7)^6$ $a = 20 \quad r = \frac{20(0.7)}{20} = 0.7$ $S_n = \frac{a(1-r^n)}{1-r}$ $S_7 = \frac{20(1-0.7^7)}{1-0.7}$ $= 61.17638$ $= 61.18 \text{ mg}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit</i> - Work of merit, for example, correct explanation in part (i) or states first term is 20 or common ratio is 0.7 <i>Mid Partial Credit:</i> - Any correct substitution into S_n formula <i>High Partial Credit:</i> - Part (i) correct and fully correct substitution into S_n formula
(a) (v)	$S_\infty = \frac{a}{1-r}$ $= \frac{20}{1-0.7}$ $= 66.\dot{6}$ $= 66.67 \text{ mg}$	Scale 5B (0, 3, 5) <i>Partial Credit</i> S_∞ with some correct substitution

Q9	Model Solution – 50 Marks	Marking Notes
(b) (i)	$k + k(0.65) + k(0.65)^2 + k(0.65)^3 + \dots + k(0.65)^6$ $a = k \quad r = \frac{k(0.65)}{k} = 0.65$ $S_n = \frac{a(1-r^n)}{1-r}$ $= \frac{k(1-0.65^n)}{1-0.65}$ $= \frac{k(1-0.65^n)}{0.35}$ $= \frac{20k(1-0.65^n)}{7}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> • Work of merit, for example, identifies a or r or two terms of series <i>Mid Partial Credit:</i> • Any correct substitution into S_n formula <i>High Partial Credit:</i> • Fully correct substitution into S_n formula
(b) (ii)	$\frac{20k(1-0.65^6)}{7} = 45$ $20k(1-0.65^6) = 45 \times 7$ $k = \frac{45 \times 7}{20(1-0.65^6)}$ $= \frac{63}{4(1-0.65^6)}$ $= 17.03474129$ $= 17 \text{ mg}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> • Work of merit, for example, sets up an equation in k <i>Mid Partial Credit:</i> • Any correct transposition to isolate k <i>High Partial Credit:</i> • Finds $k = \frac{45 \times 7}{20(1-0.65^6)}$ or equivalent

Q10	Model Solution – 50 Marks	Marking Notes
(a)	Using Pythagoras' Theorem $ SL ^2 = 8^2 + x^2$ $= 64 + x^2$ $ SL = \sqrt{64 + x^2}$ $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$ $T_1 = \frac{\sqrt{64 + x^2}}{5}$ $ LF = 20 - x$ $T_2 = \frac{20 - x}{15}$ $T(x) = T_1 + T_2$ $T(x) = \frac{\sqrt{64 + x^2}}{5} + \frac{20 - x}{15}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, attempt to use Pythagoras' Theorem or states $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$ <i>Mid Partial Credit:</i> - Finds $SL = \sqrt{64 + x^2}$ <i>High Partial Credit:</i> - Finds $SL = \sqrt{64 + x^2}$

Q10	Model Solution – 50 Marks	Marking Notes
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(b)
&
(c)

x (km)	0	2.5	5.0	7.5	10.0	12.5	15.0
Subscriptions ('000s)	2.93	2.84	2.89	3.03	3.23	3.47	3.73

The graph shows a curve representing the relationship between distance and time. The x-axis is labeled 'Distance (km)' and ranges from 0 to 15.0 with major grid lines every 2.5 units. The y-axis is labeled 'Time (hours)' and ranges from 2.0 to 4.0, with a break between 2.0 and 2.5. Major grid lines are every 0.5 units. The curve starts at (0, 2.93), dips slightly to (2.5, 2.84), rises to (5.0, 2.89), and then continues to rise more steeply through (7.5, 3.03), (10.0, 3.23), (12.5, 3.47), and ends at (15.0, 3.73).

Scale 10D (0, 3, 5, 8, 10)

Low Partial Credit:

- Two points found correctly
- One point plotted correctly

Mid Partial Credit:

- Four points plotted correctly

High Partial Credit:

- Five points plotted correctly

Full Credit – 1:

- All points plotted correctly but not joined or joined inappropriately
- Consistent incorrect rounding

Q10	Model Solution – 50 Marks	Marking Notes
(d)	$T(x) = \frac{\sqrt{64+x^2}}{5} + \frac{20-x}{15}$ $= \frac{(64+x^2)^{\frac{1}{2}}}{5} + \frac{20-x}{15}$ $T'(x) = \frac{\frac{1}{2}(64+x^2)^{-\frac{1}{2}} 2x}{5} - \frac{1}{15}$ $= \frac{x(64+x^2)^{-\frac{1}{2}}}{5} - \frac{1}{15}$ $= \frac{x}{5\sqrt{64+x^2}} - \frac{1}{15}$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, states $(64+x^2)^{\frac{1}{2}}$ <i>Mid Partial Credit:</i> - Any correct differentiation <i>High Partial Credit:</i> $T'(x) = \frac{x(64+x^2)^{-\frac{1}{2}}}{5}$
(e)	$\frac{x}{5\sqrt{64+x^2}} - \frac{1}{15} = 0$ $\frac{x}{5\sqrt{64+x^2}} = \frac{1}{15}$ $3x = \sqrt{64+x^2}$ $9x^2 = 64+x^2$ $9x^2 - x^2 = 64$ $8x^2 = 64$ $x^2 = 8$ $x = \pm 2\sqrt{2}$ $x = 2\sqrt{2} \text{ [km]} \quad x \geq 0$	Scale 10D (0, 3, 5, 8, 10) <i>Low Partial Credit:</i> - Work of merit, for example, finds $(64+x^2)^{\frac{1}{2}}$ <i>Mid Partial Credit:</i> - Any correct differentiation <i>High Partial Credit:</i> $T'(x) = \frac{x(64+x^2)^{-\frac{1}{2}}}{5}$

Q10	Model Solution – 50 Marks	Marking Notes
(f)	$T(x) = \frac{\sqrt{64+x^2}}{5} + \frac{20-x}{15}$ $T(2\sqrt{2}) = \frac{\sqrt{64+(2\sqrt{2})^2}}{5} + \frac{20-2\sqrt{2}}{15}$ $= \frac{3\sqrt{72} + 20 - 2\sqrt{2}}{15}$ $= \frac{20 + 16\sqrt{2}}{15}$ $= 2.8418278 \text{ hours}$ $= 170.509668 \text{ minutes}$ $= 171 \text{ minutes} = 2 \text{ hours } 51 \text{ minutes}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, any correct substitution into $T(x)$ <i>High Partial Credit:</i> - Finds $\frac{3\sqrt{72} + 20 - 2\sqrt{2}}{15}$ or equivalent
(g)	The use of 2 km instead of 20 km will not change the differential as it is a constant in the equation. The graph of the function is decreasing for all values till $2\sqrt{2}$, so the minimum will occur at $x = 2$. or $T(x) = \frac{\sqrt{64+x^2}}{5} + \frac{2-x}{15}$ $T'(x) = \frac{x}{5\sqrt{64+x^2}} - \frac{1}{15}$ Turning Point when $T'(x) = 0$ $\frac{x}{5\sqrt{64+x^2}} = \frac{1}{15}$ $15x = 5\sqrt{64+x^2}$ $225x^2 = 25(64+x^2)$ $225x^2 = 1800 + 25x^2$ $200x^2 = 1800$ $x^2 = \frac{1800}{200} = 9$ $x = \sqrt{9} = 3$ This is greater than $2\sqrt{2}$, so the minimum will be $x = 2$.	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, states there is no change to the differential or gives $T(x)$ or $T'(x)$ <i>High Partial Credit:</i> - States the graph is decreasing only